

PORTFOLIO

- **ABOUT ME**

I believe that an engineer with a diverse skillset is **priceless**.

It enables one to apply a knowledge base from a variety of areas and bring ideas that are innovative and tackle the problem with ingenuity.

In my professional experience, I have built machines ranging from UAVs to Robotic Arms to Medical ventilators. Each product that has come into realization has had intricate analysis done on it, be FEM or CFD or topological optimization.

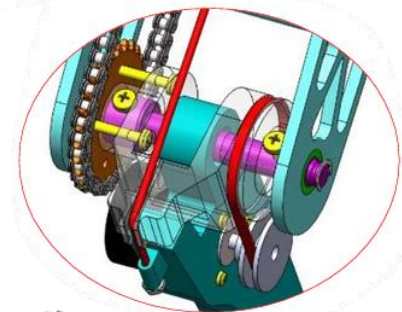
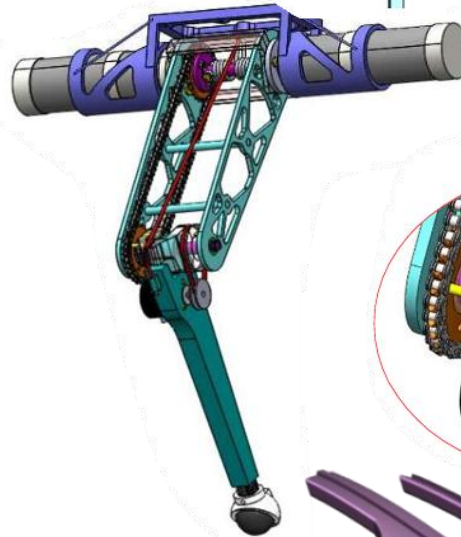
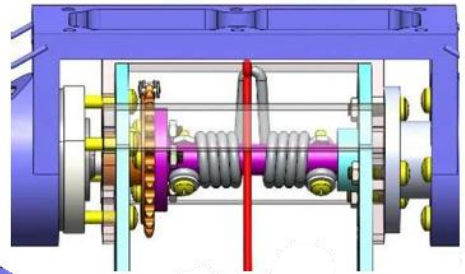
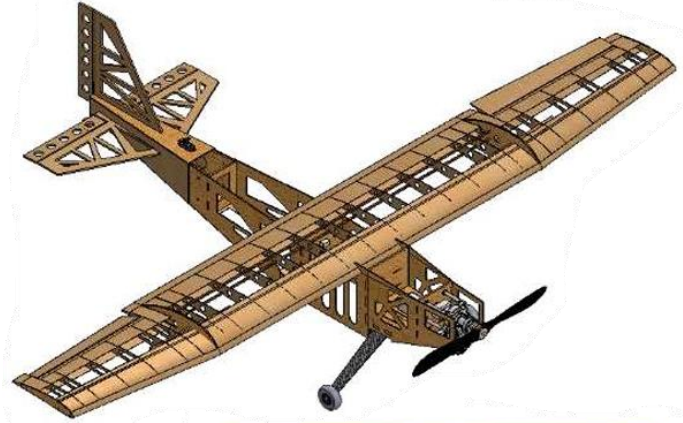
My career has also enabled me to have experience in Machine-Learning, AI, coding with numerous languages (C++, python, ROS, etc) and other software-based skills such as http, css and php.

Finally, my skills range into the complex world of fabrication, which includes CNC machining, lathe and milling, 3D printing, Glass-fiber and Carbon-fiber production.

I combine all my skills and expertise to conquer problems that are more challenging for someone with a singular experience.

- **CAD (Solidworks, NX Siemens, Fusion 360, PTC Creo)**

I have been designing complex products in CAD software for the past **8 years**. These include quadruped robots, UAVs, medical machines, prosthetics, daily products and industrial machines.



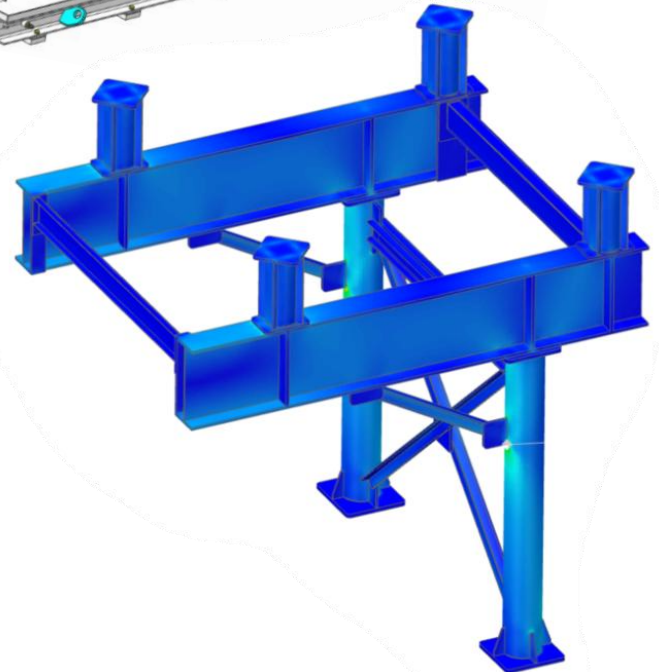
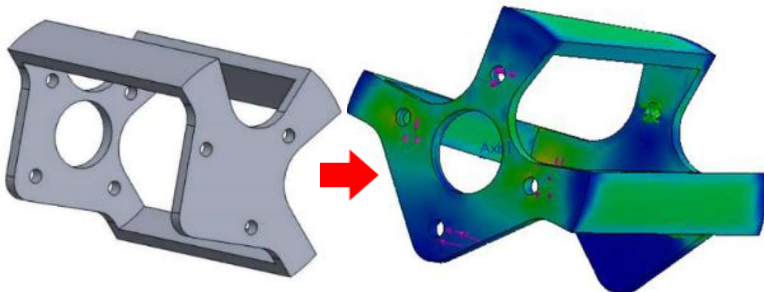
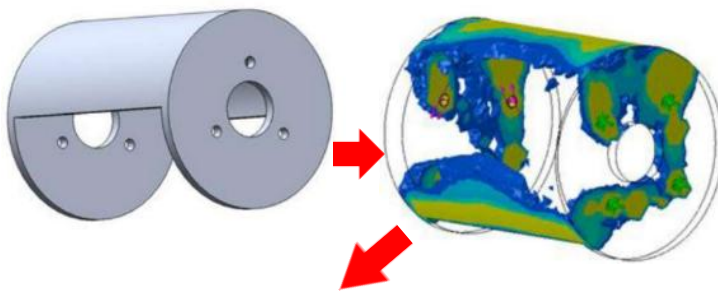
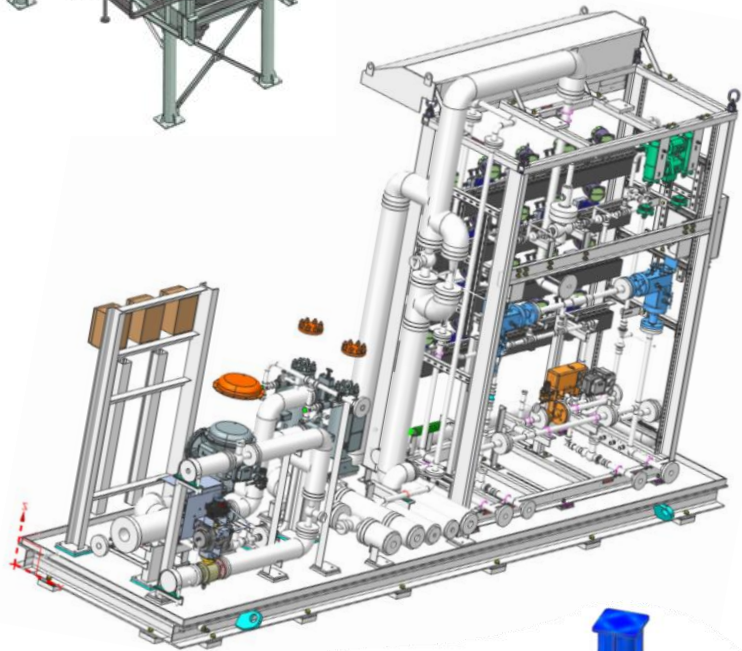
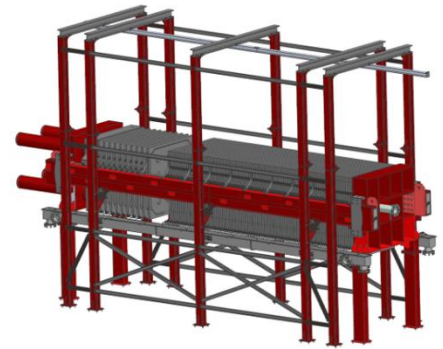
My experience of designing revolves around solidworks and NX Siemens, but I have also ventured into other software such as PTC Creo, AnSys spaceclaim, sketchup and highly detailed renders in Blender.

- **FEM and CFD (NX Siemens, Solidworks, Ansys)**

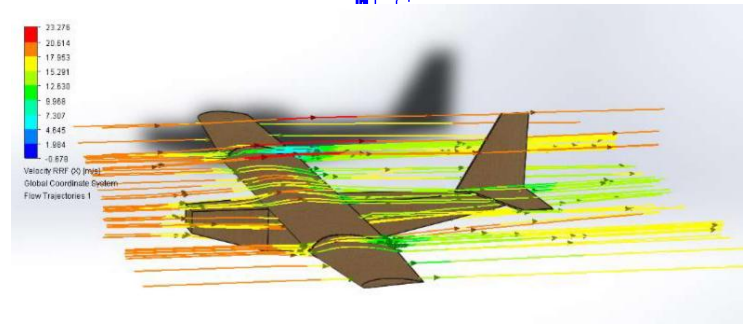
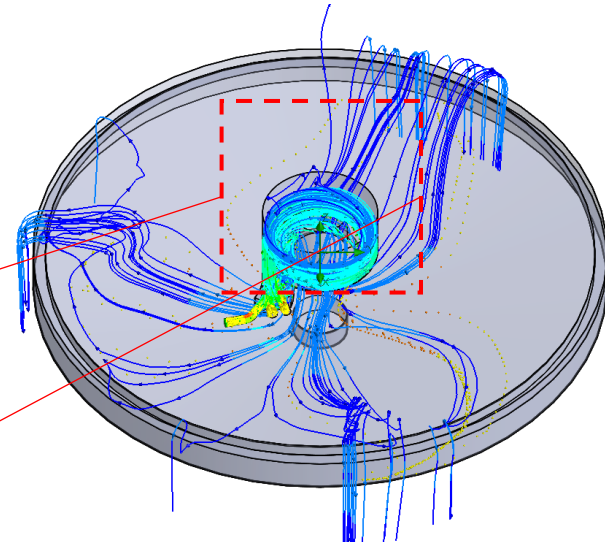
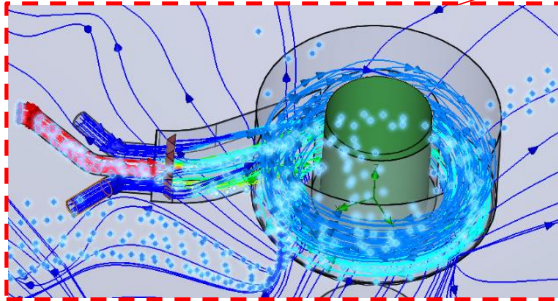
To perform Finite element Analysis for structures that do not require immense precision, I used NX Siemens or SolidWorks.

This would include testing the poisson's ratio for a rubber wheel while under stress. A silo, filled with slurry, undergoing a structural safety test. To test the wing strength of an airplane during less-than-optimal conditions and asses a FOS.

FEM has also been used to do topology optimization, to find the important force vectors and trajectories during loading, so that only the desired material is retained. This helps in reducing the material weight without affecting its strength.



CFD has frequently been used in my projects to evaluate the fluid flow through a thickener to examine if the particles overflow or not, or if pure water is being transported, or if proper mixing is taking place.

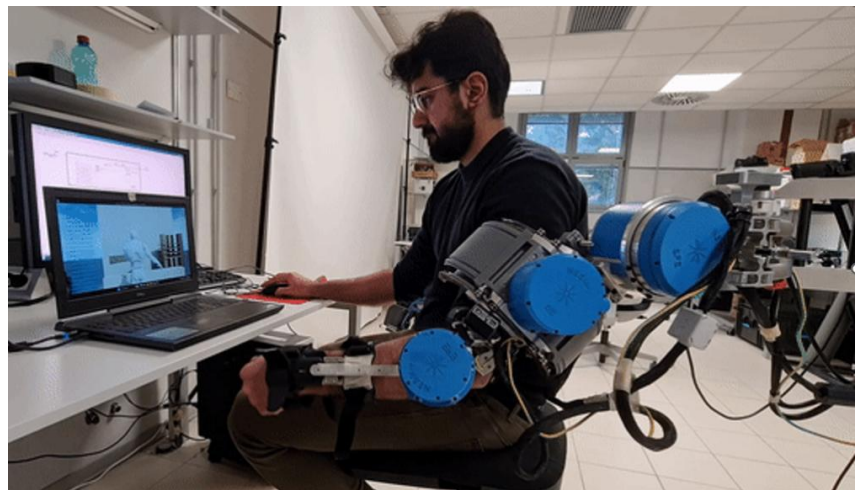
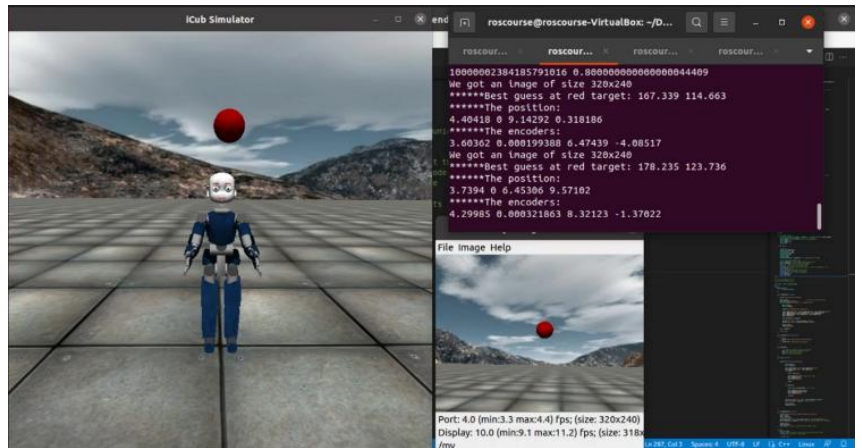


• ROBOTICS AND CONTROL

I have worked with ROS and teleoperation of robots (ICub robot) by opening serial ports and using them to control the limbs of the mobile robot remotely. Furthermore, the eye sockets were automated through color correction, to follow a ball and its trajectory.

In my Bachelor's Final year project, I designed a single leg of a quadruped robot with an innovative SEA design, which reduces energy input required to complete a stride. It was operated through BLDC motors and a PID control made in Matlab. The control was implemented through LabView GUI alongside the multiple sensors on-board the robot for capable gaits.

In my master's thesis, I had the opportunity to work with the NESM Gamma exoskeleton to enable its Torque/Position control system to collaborate with a virtual reality engine (Unreal Engine) through inverse kinematics and a NI MyRio controller. It enabled the user to experience haptic feedback from the virtual objects.



• FABRICATION

Throughout my years as a mechanical engineer, I have learnt multiple manufacturing techniques. In addition, continue to update them as new technologies are developed.

Whether it be custom torsional springs or custom size gears for a transmission or lightweight aluminum weight bearing structure, I have designed them and fabricated using CNC, lathe, Sheet Metal, Tube Forming and Milling machines.

I have also worked additive manufacturing, making prosthetics, fiberglass airfoils, carbon fiber structures and other custom products using 3D printing (Prusa, BambuLab) with diverse materials such as carbon, PLA and ABS.



B

